BIOCHEMISTRY CUMULATIVE EXAM

February 11, 2000

Molecular Biology

1. (40 points) Define each of the following terms.
   A. Restriction enzyme.
   B. Polymerase chain reaction.
   C. Topoisomerase.
   D. Site-directed mutagenesis.
   E. Frameshift mutation.
   F. Initiation codon.
   G. Nucleic acid splicing.
   H. Genomic library.
   I. cDNA library.
   J. Transfer RNA.

2. A sample of a small DNA molecule (molecular weight 3,100,000) is purified to homogeneity, so that no chemical or sequence contaminants are present. Yet, the DNA sample is found to give three bands when subjected to electrophoresis through a gel of agarose.

   A. (10 pts) Why are three bands observed?

   B. (5 pts) After the DNA sample is treated with Bam H1 and tested again by electrophoresis, only one band is observed. Why is only one band now observed?

   C. (10 pts) List and justify any additional facts that you can conclude about the nature of the original DNA.

3. Five microgram samples of *E. coli* DNA and human DNA are each digested to an average size of 8,000 nucleotide pairs. Each sample is then heated (in aqueous solution) to 98° C for 10 minutes and then very slowly cooled to 38° C over a period of 5 days. During the cooling process, the absorbance at 260 nm is recorded continuously for each sample.

   A. (5 points) Why is a wavelength of 260 nm chosen?
   B. (30 points) What important differences will be observed in the results for the *E. coli* and human DNA samples? Why?